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by

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DEPARTMENT OF COMPUTER SCIENCE

**VEERANARI CHAKALI ILAMMA WOMEN'S
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**DEPARTMENT OF COMPUTER SCIENCE
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This is to certify that the Project work entitled “”CUSTOMER RETENTION AI”” submitted by **KHUSHI KUMARI** bearing **HT. NO: 101324504007** in partial fulfilment for the award of Master of Science in Computer Science to the Veeranari Chakali Ilamma Women's University, Hyderabad is a record of Bonafide work carried out by her under the Guidance of **Dr. S. NARSIMHA RAO**, Assistant Professor, Department of Computer Science, Veeranari Chakali Ilamma Women's University, Hyderabad during the Academic year August-2026.

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DECLARATION

I hereby declare that the work described in this project entitled “**CUSTOMER RETENTION AI**” which is being submitted by me in partial fulfilment for the award in the Department of Computer Science, Veeranari Chakali Ilamma Women's University, Hyderabad(T.S), in partial fulfilment for the award of degree of **Master of Science in Computer science** is the result of investigations carried out by me under the guidance of Dr. S. NARSIMHA RAO ,Assistant Professor, Department of Computer Science ,VCIWU, Hyderabad.

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ABSTRACT

Customer churn is one of the most pressing challenges facing subscription-based telecommunications providers, where the cost of acquiring a new customer typically far exceeds the cost of retaining an existing one. This Project presents CustomerRetention AI, an end-to-end machine learning system designed to predict customer churn, diagnose the underlying causes of that churn, categorize customers by retention and purchase potential, and recommend the most suitable subscription plan to retain at-risk or upsell-ready customers. The system was built on a real-world telecom dataset of 6,418 customer records containing demographic, service-usage, billing, and churn-outcome attributes. The project follows the complete data science lifecycle — business understanding, data cleaning, feature engineering, exploratory data analysis, multi-algorithm model training and selection— culminating in an interactive Streamlit web application.

Three machine learning algorithms (Logistic Regression, Random Forest, and XGBoost) were trained and compared for the binary churn-prediction task. Random Forest was selected as the best-performing model, achieving an ROC-AUC of 0.894, an accuracy of 83.78%, a precision of 70.11%, a recall of 76.37%, and an F1-score of 0.731 on the held-out test set. A second set of three algorithms (Logistic Regression, Decision Tree, and Random Forest) was trained for multiclass customer categorization into High-Risk, Medium-Risk, Loyal, and Premium-Ready segments, with Random Forest again selected as the best model at 81.15% accuracy. A diagnostic churn-reason classifier, also based on Random Forest and trained exclusively on churned customers, was built to explain why a given customer is likely to leave, achieving 34.58% accuracy across five overlapping reason categories. A custom-built subscription recommendation engine, trained on a hand-crafted catalogue of subscription plans mapped to customer complaint and need patterns, uses a single Random Forest algorithm evaluated through five-fold cross-validated classification accuracy (36.54%), regression mean-squared-error (57.97) and R-squared (0.814), and a Chi-Squared test of statistical association ($\chi^2 = 676.0$, $p < 0.001$).

The resulting web application integrates all four capabilities — churn prediction and diagnosis, customer categorization, subscription recommendation, and an interactive Plotly-based dashboard — into a single, presentable tool intended to support telecom retention and marketing teams in reducing churn and maximizing customer lifetime value. This report documents the complete methodology, the comparative performance of every algorithm evaluated, and the reasoning behind each model-selection decision, with particular emphasis on why Random Forest consistently outperformed its alternatives across both the churn-prediction and customer-categorization tasks.

Keywords: *Customer Churn Prediction, Random Forest, XGBoost, Logistic Regression, Customer Segmentation, Recommendation System, Streamlit, Telecom Analytics, Machine Learning Algorithm.*

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The telecommunications industry operates in a highly saturated and competitive market where customers can switch providers with minimal friction, a phenomenon commonly referred to as churn. Industry research consistently shows that acquiring a new customer costs five to seven times more than retaining an existing one, making churn management a strategic priority rather than a peripheral operational concern. With the proliferation of digital services, telecom companies now generate vast amounts of structured data — covering demographics, service subscriptions, billing history, and support interactions — that can be leveraged through machine learning to anticipate and prevent customer attrition before it occurs.

Traditional churn-management approaches have relied on reactive strategies, such as offering discounts only after a cancellation request is received. This project instead adopts a proactive, data-driven approach: using historical customer data to build predictive models that flag at-risk customers in advance, diagnose the specific reasons behind their risk, and recommend a personalized retention action — namely, an appropriately tailored subscription plan — before the customer decides to leave. This shift from reactive to proactive retention management mirrors a broader trend across subscription-based industries, where predictive analytics is increasingly viewed not as an optional enhancement but as a core operational capability.

The growing availability of open-source machine learning tooling — including Scikit-learn, XGBoost, and lightweight frameworks such as Streamlit — has also lowered the barrier to building and shipping production-quality predictive systems, making it feasible for a project of this scope to progress from raw data to a fully interactive web application within an academic timeframe.

However, accurately identifying customers who are likely to churn is only one part of the retention problem. From a business perspective, a prediction that a customer is likely to leave provides limited value unless it is accompanied by meaningful insight into the factors contributing to that risk and a practical action that can be taken in response. For example, churn may be associated with high monthly charges, poor service satisfaction, contract characteristics, low engagement, or unresolved customer complaints. Therefore, an effective customer-retention system should move beyond simple binary prediction and provide interpretable insights that help decision-makers understand the potential drivers of customer attrition.

Another important challenge is that customers do not represent a homogeneous group. Some customers may be at immediate risk of leaving, while others may be loyal but have strong potential

for premium services or additional purchases. Treating all customers with the same retention strategy can result in inefficient use of marketing resources and unnecessary discounts. Customer categorization and segmentation can therefore help organizations distinguish between different behavioural and business-value groups, allowing retention efforts, promotional campaigns, and subscription recommendations to be targeted more effectively.

In addition, the usefulness of a machine learning solution depends not only on model performance but also on its ability to be translated into an accessible decision-support system. A highly accurate predictive model has limited practical value if business users cannot easily interact with it or interpret its results. For this reason, there is increasing emphasis on integrating predictive models with dashboards and interactive applications that present customer-level insights, risk scores, diagnostic information, and recommended actions in a clear and usable format.

Against this background, **CustomerRetention AI** was developed as an integrated machine learning-based customer retention system for the telecommunications domain. The project combines churn prediction, churn-reason analysis, customer categorization, subscription-plan recommendation, and interactive business visualization within a single application. Rather than treating these tasks as isolated analytical exercises, the system is designed to support a complete decision-making workflow: identifying customers who may leave, understanding their likely risk factors, categorizing their retention or purchase potential, and recommending an appropriate action.

The overall objective is therefore to demonstrate how data-driven techniques can be transformed from standalone predictive models into a more comprehensive and actionable customer-retention solution. By combining machine learning with an interactive layer, the project aims to bridge the gap between analytical model development and practical business use, illustrating how customer data can support more proactive, targeted, and informed retention strategies.

1.2 Problem Statement

Telecom providers struggle to answer three interconnected questions in a timely and actionable manner:

- (1) Which customers are likely to churn in the near future?
- (2) Why are these customers at risk — is it price sensitivity, poor service quality, a competitor offer, or general dissatisfaction?
- (3) What can be offered to these customers, or to loyal high-value customers, to improve retention and revenue?

Existing systems in many organizations address these questions in isolation, using separate spreadsheets, dashboards, or ad-hoc analyses that are not integrated into a single decision-support tool. This fragmentation slows down retention response time and reduces the effectiveness of interventions, since an agent working from a churn score alone has no direct guidance on what action to take next.

1.3 Objectives of the Study

1. To clean, preprocess, and engineer meaningful features from a raw telecom customer dataset of 6,418 records.
2. To build and rigorously compare multiple machine learning algorithms for binary churn prediction, and to select the best-performing model using accuracy, precision, recall, F1-score, and ROC-AUC as joint evaluation criteria.
3. To design a diagnostic framework that explains the likely reason behind a customer's predicted churn risk, using both model-based feature importance and a customer-specific rule-based narrative.
4. To construct a multiclass customer categorization model that segments customers by retention and purchase potential, and to compare algorithm performance in the same rigorous manner as the churn task.
5. To design and train a subscription recommendation engine using a custom-built plan catalogue, evaluated with classification accuracy, regression error metrics, and a statistical test of association.
6. To develop an interactive dashboard that visualizes churn, revenue, and subscription trends across the customer base.
7. To integrate all components into a single, Streamlit web application with a coherent, business-usable interface.

1.4 Scope of the Project

The scope of this project covers the complete machine learning lifecycle applied to a single telecom customer dataset of 6,418 records and 32 original attributes. It includes data cleaning and feature engineering, exploratory data analysis, training and evaluating multiple algorithms for two distinct classification problems (churn prediction and customer categorization), building a recommendation engine on a purpose-built dataset, and all components through a single web interface. The project does not cover real-time data streaming, integration with live production billing systems, or automated campaign execution — these are identified as future extensions in

Section 16. The project is also intentionally scoped to a single static dataset snapshot rather than a continuously updated data pipeline, consistent with its nature as an academic exercise.

The scope further includes comparative evaluation of multiple machine learning algorithms to determine the most suitable model for each classification task based on relevant performance metrics. The project also incorporates diagnostic analysis to provide insights into the factors associated with customer churn, thereby extending the system beyond a simple prediction-based approach. Customer-level categorization is included to distinguish between different behavioural and retention-oriented groups, enabling the system to support more targeted business decisions.

The recommendation component is limited to suggesting subscription plans based on the patterns and relationships represented within the purpose-built recommendation dataset. It is intended to demonstrate a data-driven approach to matching customer needs or complaints with potentially suitable retention offers and does not represent a live commercial recommendation platform. Similarly, the web application is designed as an interactive prototype that demonstrates the integration of the trained models, analytical outputs, and business-oriented visualizations within a unified interface.

Model performance is evaluated using appropriate classification and analytical metrics, with the objective of comparing alternative approaches and selecting models that provide useful predictive performance and interpretable outputs. However, the results are limited to the dataset and experimental conditions used in this study and should not be interpreted as a guarantee of identical performance across different telecom companies, customer populations, or real-world operational environments. Despite these limitations, the project provides a complete end-to-end demonstration of how machine learning techniques can be applied to support proactive customer retention and data-driven decision-making.

1.5 Significance of the Study

This project demonstrates how a moderately sized dataset can be transformed into a complete, business-ready decision-support tool using standard, reproducible machine learning practices. Beyond its academic value as an exercise in applied data science, the system offers practical significance: it consolidates churn prediction, diagnosis, categorization, and recommendation into a single interface, reducing the analytical overhead typically required to act on customer risk signals. For a retention team, the ability to see not only who is likely to churn but why, and what to offer instead, represents a meaningful improvement over siloed reporting tools. The comparative model-selection methodology used throughout this project — evaluating multiple algorithms against multiple metrics rather than relying on a single score — is also intended to serve as a template for similarly structured classification problems in other domains.

CHAPTER 2

BUSINESS UNDERSTANDING

2.1 Business Objective

The core business objective is to reduce customer churn and protect recurring revenue by equipping the retention team with a predictive, explainable, and prescriptive tool. Rather than simply reporting historical churn rates, the system is designed to look forward — identifying which currently active customers are at elevated risk, explaining the dominant driver of that risk, and recommending a concrete next action (a specific subscription plan) that addresses the customer's underlying need. This objective directly shaped every downstream design decision in the project, from the choice of business-rule-derived category labels to the structure of the recommendation catalogue.

2.2 Churn Problem in Subscription Business

In subscription-based businesses such as telecom, revenue is a function of the number of active subscribers and their average revenue per user, both of which are directly eroded by churn. Because subscription revenue is recurring, even a small monthly churn rate compounds significantly over a year — a 5% monthly churn rate, for example, implies that more than half of a customer base would be lost within twelve months if unaddressed. Understanding not just the rate but the composition of churn (which segments, which reasons, which contract types) is essential for designing effective, targeted retention programs rather than blanket discounting strategies that erode margins.

The dataset used in this project reflects this dynamic clearly: with an overall churn rate of 27.0% across 6,418 customers, and Month-to-Month contract holders churning at a substantially higher rate than customers on annual or biennial contracts, contract structure alone emerges as a first-order lever available to the business, well before any individual predictive model is even considered.

Churn also creates costs that go beyond the immediate loss of subscription revenue. When an existing customer leaves, the company may need to spend additional resources on marketing, promotions, and acquisition activities to replace that customer. At the same time, customers who have remained with the company for a longer period may represent greater lifetime value through continued subscription payments and the potential purchase of additional services. Therefore, identifying customers who are at risk before they actually leave can help businesses prioritize retention efforts and allocate resources more effectively.

Another important aspect of the churn problem is that not all customers require the same retention strategy. A customer considering leaving because of high monthly charges may require a different intervention from a customer experiencing service-related issues or dissatisfaction with their current plan. Similarly, some customers may not be at immediate risk of churn but could be suitable for premium plans or additional services. This makes customer-level analysis important, as a single blanket strategy may result in unnecessary discounts and reduced profitability.

For this reason, the churn problem in subscription businesses can be viewed as more than simply predicting whether a customer will leave. An effective retention approach should also help the business understand the characteristics associated with churn and support decisions about which customers should be prioritized and what type of action may be appropriate. These considerations form the basis for the subsequent predictive modelling, customer categorization, diagnostic analysis, and recommendation components developed in this project.

2.3 Stakeholders and Use Case

The primary stakeholders for this system are the Customer Retention / Customer Experience team, who would use the churn prediction and diagnostic sections to prioritize outreach; the Marketing and Product teams, who would use the categorization and recommendation sections to design targeted campaigns and plan bundles; and business leadership, who would use the interactive dashboard for monitoring churn KPIs, revenue trends, and subscription performance at a glance. A typical use case involves a retention agent looking up a customer flagged as high-risk, reviewing the predicted churn reason, and immediately presenting the recommended subscription plan as a retention offer during a support call.

2.4 Success Criteria

- Churn prediction model achieving an ROC-AUC of at least 0.85 on held-out test data.
- Customer categorization model achieving at least 75% classification accuracy.
- A statistically significant relationship between customer complaint/need and the recommended subscription plan.
- A fully functional, integrated web application covering all four required capabilities.
- Clear, business-interpretable outputs (probabilities, categories, and explanations) rather than opaque model scores.

CHAPTER 3

LITERATURE REVIEW

3.1 Overview of Customer Churn Prediction

Customer churn prediction has been extensively studied in the telecommunications, banking, and subscription-service sectors. Early approaches relied on statistical methods such as logistic regression and survival analysis to estimate the probability of a customer discontinuing service within a given time window. Logistic regression remains a popular baseline because its coefficients are directly interpretable as log-odds contributions of each feature, making it easy for business stakeholders to understand which factors increase or decrease churn risk. However, its assumption of a linear decision boundary in the log-odds space limits its ability to capture complex, non-linear interactions between features.

With the growth of computational power and structured customer data, ensemble tree-based methods — including Random Forest and gradient boosting variants such as XGBoost — have become the standard approach in both academic literature and industry practice, consistently outperforming single linear models on tabular customer data due to their ability to capture non-linear interactions between features such as tenure, contract type, and billing behavior. Random Forest, introduced by Breiman (2001), builds an ensemble of decision trees trained on bootstrap-resampled subsets of the data and random feature subsets, averaging their predictions to reduce variance and overfitting relative to a single decision tree. XGBoost, introduced by Chen and Guestrin (2016), instead builds trees sequentially, with each new tree trained to correct the residual errors of the ensemble so far, often achieving superior predictive accuracy at the cost of additional hyperparameter sensitivity and reduced interpretability relative to Random Forest.

Timing is also an important factor in churn management. Once a customer has already decided to leave or completed the cancellation process, the opportunity for successful retention may be significantly reduced. Early identification of potential churn allows businesses to intervene before the customer's decision becomes final. However, since retention resources such as discounts, personalized offers, and customer-support interventions may be limited, businesses must also prioritize customers based on their level of risk and potential value rather than applying the same level of effort to every customer.

In addition, analysing churn patterns at an aggregate level can help businesses identify broader operational issues affecting multiple customers. For example, consistently high churn within a particular contract type, service category, or customer group may indicate an underlying issue that requires a broader business response rather than individual customer-level intervention.

Combining overall churn analysis with individual predictions can therefore support both strategic decision-making and targeted customer retention efforts.

Ultimately, the objective is not simply to reduce the churn percentage at any cost, but to improve retention in a way that is both effective and economically practical. A useful churn-management system should help decision-makers balance the cost of retention activities against the potential value of retaining a customer, enabling more focused and data-driven subscription business strategies.

Machine Learning in Customer Churn Prediction

Customer churn prediction is commonly formulated as a supervised machine learning classification problem, in which historical customer records are used to learn patterns associated with customer departure. Each customer is represented through a set of input features, such as demographic information, service usage, contract details, billing behaviour, and customer interactions, while the target variable indicates whether the customer churned or remained with the company. Once trained, the resulting model can estimate the probability that an unseen customer belongs to the churn class.

The effectiveness of churn prediction depends not only on the choice of algorithm but also on the quality and relevance of the available features. Customer churn is influenced by multiple interacting factors, including pricing, service quality, contract commitment, tenure, and competitive alternatives. Machine learning models are particularly useful in this context because they can identify complex relationships between these variables that may not be apparent through simple descriptive analysis

3.2 Existing Retention Strategies

Traditional retention strategies in telecom have historically been reactive, triggered only when a customer initiates a cancellation request (a save-desk model) or applied uniformly across a broad customer segment (blanket discounting). More sophisticated approaches described in industry and academic literature use churn scores to trigger proactive outreach before the customer expresses intent to leave, often paired with personalized offers rather than generic discounts. This project builds directly on that proactive philosophy, extending it further by pairing the churn score with a diagnostic explanation and a concrete, personalized subscription recommendation, rather than stopping at the prediction stage as many published churn-prediction studies do.

3.3 Customer Segmentation Methods

Customer segmentation approaches in the literature range from unsupervised clustering methods (such as k-means or RFM — Recency, Frequency, Monetary — analysis) to supervised classification against business-defined categories. Unsupervised methods are useful for discovering unknown structure in the data but produce clusters that are not inherently tied to actionable business labels, requiring a separate interpretation step to map each cluster to a business action. This project instead adopts a hybrid approach: business-rule-derived category labels (High-Risk, Medium-Risk, Loyal, Premium-Ready) are constructed from domain knowledge and then used as supervised targets for classification models, combining the interpretability of business rules with the generalization capability of machine learning, so that the resulting model can categorize new or unseen customers without requiring the original rule engine to be re-run.

One advantage of using business-defined categories is that the resulting customer groups can be directly connected to potential business strategies. For example, High-Risk customers may require immediate retention attention, while Medium-Risk customers can be monitored and targeted through preventive engagement strategies. Loyal customers represent a relatively stable customer base that should be maintained through continued service quality and relationship-building efforts, whereas Premium-Ready customers may represent opportunities for upselling or recommending higher-value subscription plans. This direct connection between categories and potential actions improves the practical usefulness of the segmentation process.

3.4 Recommendation Systems in Retention

Recommendation systems are widely used in e-commerce and media streaming to suggest products or content based on user behaviour, typically through collaborative filtering (which leverages patterns across many users) or content-based filtering (which leverages the attributes of the items and the user themselves). Their application to retention-oriented subscription-plan recommendation is comparatively less common in academic literature but conceptually analogous: instead of recommending a product based on purchase history, the system recommends a subscription plan based on a customer's stated or inferred complaint and need profile, which is a content-based filtering approach using customer attributes rather than collaborative signals from other users. This choice was deliberate, since a cold-start customer with no purchase history can still receive a meaningful recommendation as long as their complaint or need profile is known.

3.5 Research Gap

While churn prediction, customer segmentation, and recommendation systems are each well studied independently, relatively few published systems integrate all three into a single, explainable tool aimed specifically at telecom retention teams. Most academic work focuses narrowly on maximizing a single predictive metric (such as AUC) without addressing the downstream question of what action to take once a customer is flagged as high-risk. This project addresses that gap by combining prediction, diagnosis, categorization, and prescriptive recommendation into one coherent, end-to-end pipeline with a functional user interface, and by explicitly documenting the comparative performance of every candidate algorithm rather than reporting only the final selected model.

An additional limitation in many existing approaches is the separation between analytical model development and practical usability. Even when accurate predictive models are developed, the results are often presented only through experimental metrics, tables, or research-based analysis rather than through an interface that allows non-technical users to interact with the system and interpret its outputs. By integrating the analytical components within a single interactive application, this project attempts to bridge the gap between model performance and practical decision support, demonstrating how predictive insights can be translated into understandable customer-level information and actionable retention recommendations.

CHAPTER 4

DATASET DESCRIPTION

4.1 Dataset source

The dataset used in this project is a telecom customer dataset comprising 6,418 customer records and 32 original attributes, covering customer demographics, service subscriptions, contract and billing details, and churn outcome labels. Each row represents a unique customer identified by a Customer_ID, and the dataset reflects a realistic mix of active, newly joined, and churned customers, making it suitable for both binary churn classification and diagnostic churn-reason analysis. The dataset was supplied in comma-separated-value (CSV) format and loaded into a Pandas DataFrame for all subsequent processing.

4.2 Attribute Description

The dataset attributes fall into four broad groups, summarized in Table 4.1:

Group	Representative Attributes	Count
Demographic	Gender, Age, Married, State, Number_of_Referrals	5
Service	Phone_Service, Multiple_Lines, Internet_Service, Internet_Type, Online_Security, Online_Backup, Device_Protection_Plan, Premium_Support, Streaming_TV, Streaming_Movies, Streaming_Music, Unlimited_Data	12
Billing & Contract	Contract, Paperless_Billing, Payment_Method, Monthly_Charge, Total_Charges, Total_Refunds, Total_Extra_Data_Charges, Total_Long_Distance_Charges, Total_Revenue, Tenure_in_Months, Value_Deal	11
Outcome	Customer_Status, Churn_Category, Churn_Reason, Customer_ID	4

Table 4.1 — Dataset attribute grouping

4.3 Target Variables

Two supervised target variables were engineered from the raw Customer_Status field and business rules: Churn_Label, a binary indicator (1 = Churned, 0 = Stayed or Joined) used for the churn prediction task, and Customer_Category, a four-class label (High-Risk, Medium-Risk, Loyal,

Premium-Ready) derived from a combination of churn status, contract length, tenure, and revenue, used for the categorization task. In addition, the existing Churn_Category field (Competitor, Attitude, Dissatisfaction, Price, Other) present only for churned customers was used as the target for the diagnostic churn-reason model. This dual use of the term “category” — once for the engineered retention-potential segment and once for the pre-existing churn-reason label — is disambiguated throughout this report by consistently referring to the former as Customer_Category and the latter as Churn_Category.

4.4 Data Quality Issues

Initial data profiling revealed several data quality issues that required treatment before modeling: missing values in Value_Deal (3,548 records), Multiple_Lines (622 records), and eight internet-dependent add-on service columns (1,390 records each, corresponding to customers without an internet subscription); a small number of negative Monthly_Charge values indicating data-entry errors; and Churn_Category / Churn_Reason fields that were populated only for the subset of customers with Customer_Status equal to 'Churned', by design. These issues are addressed systematically in Section 5.

the cleaning process preserved all 6,418 customer records, with no duplicate rows identified in either version. The 107 negative values in Monthly_Charge present in the original dataset were corrected during cleaning, and missing categorical values were replaced using contextually meaningful categories rather than treating all missing values as ordinary unknown observations. For example, the 3,548 missing Value_Deal entries were represented as “No Deal,” while customers without internet service were explicitly identified through the Internet_Type category “No Internet,” ensuring that the absence of a service was retained as meaningful information rather than incorrectly imputed as a standard service value.

The cleaned dataset also expanded the original 32 attributes to 45 attributes through the creation of additional modelling and analytical features. These included a binary Churn_Label, tenure grouping, total add-on service count, revenue and charge-related measures, contract and spending indicators, referral and senior-customer flags, and the business-defined Customer_Category. All newly created features contained no missing values. The Churn_Category and Churn_Reason fields were intentionally retained with missing values for non-churned customers because these variables are only applicable to customers who had actually churned; treating those missing values as conventional data-quality errors would have altered the meaning of the original dataset.

CHAPTER 5

DATA PREPROCESSING

5.1 Data Cleaning

Data cleaning was performed using the Pandas library in Python. The cleaning process focused on correcting invalid values, treating missing data based on business logic rather than blanket statistical imputation, and ensuring that every downstream feature was well-defined for every customer record. This business-logic-first philosophy was chosen deliberately: applying a generic imputation strategy (such as filling every missing value with the column mean or mode) would have obscured the structural reason behind each missing value, and in several cases would have actively introduced incorrect information into the dataset, as explained in the following subsection.

In addition to missing-value treatment, the cleaning process included validation of numerical fields to identify values that were outside their meaningful business range. Negative values identified in the `Monthly_Charge` attribute were treated as data-entry errors and corrected before further analysis, since a negative recurring subscription charge does not represent a valid customer billing value in the context of this dataset. The dataset was also checked for duplicate customer records and consistency across key customer attributes. These steps ensured that the cleaned dataset retained the original customer population while improving the reliability and usability of the data for subsequent exploratory analysis, feature engineering, and model development.

Importance of Data Quality

Data quality is a fundamental requirement for reliable machine learning because predictive models learn directly from the information provided during training. Missing values, invalid entries, inconsistent categories, and incorrect data types can introduce noise or bias into the learning process and may reduce the reliability of the resulting predictions. Therefore, data preprocessing aims to transform raw customer records into a consistent and meaningful representation before feature engineering and model development.

5.2 Missing Value Treatment

Missing values were treated according to their business meaning rather than through indiscriminate mean or mode imputation. Missing `Value_Deal` entries were interpreted as the customer not being enrolled in any promotional deal and encoded as 'No Deal'. Missing `Multiple_Lines` entries were treated as 'No', consistent with customers who do not have an active phone-line add-on. The eight

internet-dependent service columns (Online_Security, Online_Backup, Device_Protection_Plan, Premium_Support, Streaming_TV, Streaming_Movies, Streaming_Music, Unlimited_Data) were missing precisely for customers without an internet subscription; these were encoded as 'No' (or 'No Internet' for Internet_Type), reflecting a structural rather than random missingness pattern. This distinction between missing-completely-at-random data and structurally missing data is a central concept in modern data-cleaning practice, and treating the two categories identically would have been methodologically incorrect.

In addition to missing-value treatment, the cleaning process included validation of numerical fields to identify values that were outside their meaningful business range. Negative values identified in the Monthly_Charge attribute were treated as data-entry errors and corrected before further analysis, since a negative recurring subscription charge does not represent a valid customer billing value in the context of this dataset. The dataset was also checked for duplicate customer records and consistency across key customer attributes. These steps ensured that the cleaned dataset retained the original customer population while improving the reliability and usability of the data for subsequent exploratory analysis, feature engineering, and model development.

5.3 Duplicate Removal

Records were checked for duplication on the Customer_ID key using Pandas' drop_duplicates function. No duplicate customer records were found in the source dataset, confirming that each row represents a unique customer and that no further de-duplication logic was required.

5.4 Outlier Handling

A small number of records contained a negative Monthly_Charge value, which is not physically meaningful and was treated as a data-entry error; these values were replaced with the column median, chosen over the mean for its robustness to the very outliers being corrected. Total_Charges, Total_Refunds, Total_Extra_Data_Charges, Total_Long_Distance_Charges, and Total_Revenue were clipped at a lower bound of zero to remove any residual negative values arising from refund adjustments, since none of these quantities can be legitimately negative in a billing context.

The outlier treatment was therefore limited to values that were clearly inconsistent with the business meaning of the corresponding attribute, rather than applying aggressive statistical removal to all extreme observations. High values in revenue, tenure, or usage-related fields were retained because they may represent genuinely valuable or high-usage customers and could contain important information for churn prediction and customer categorization. This approach ensured

that only invalid or logically impossible values were corrected while preserving meaningful variation within the customer population.

5.5 Data Type Conversion

Numeric fields such as Age, Tenure_in_Months, Monthly_Charge, and the revenue-related columns were validated and cast to appropriate integer or floating-point types. Categorical fields such as Contract, Payment_Method, and the service add-on columns were retained as string/object types for subsequent encoding, ensuring that no numeric-looking categorical code (for example, a State abbreviation) was inadvertently treated as a continuous numeric feature.

Data type validation also ensured consistency between the original dataset and the cleaned dataset before feature engineering and model preparation. Numeric variables were checked to confirm that mathematical operations and statistical calculations could be performed correctly, while categorical attributes were preserved as discrete labels representing customer characteristics, service choices, or subscription behaviour. Maintaining appropriate data types at this stage reduced the risk of preprocessing errors and ensured that subsequent encoding and modelling steps were applied to the correct set of variables.

5.6 Encoding of Categorical Variables

All categorical variables used as model inputs were transformed using Label Encoding, implemented via Scikit-learn's LabelEncoder, fitted separately for each model pipeline (churn, categorization, and recommendation) to ensure that only classes observed during training are used at inference time, with unseen categories safely mapped to a default class to avoid runtime errors in the prototypical web application. Label Encoding was preferred over One-Hot Encoding in this project because the tree-based models used throughout (Random Forest, Decision Tree, XGBoost) are able to split on integer-encoded categorical features without assuming a spurious ordinal relationship, and because One-Hot Encoding would have substantially increased the dimensionality of the feature space given the number of categorical columns with more than two levels.

The encoding process was applied consistently across the relevant modelling tasks so that the transformation of customer attributes remained compatible between model training and prediction. The fitted encoders were retained alongside the corresponding trained models and preprocessing components for use during inference in the web application. This ensured that categorical inputs supplied for new customers could be transformed into the same numerical representation expected by the trained model, supporting reliable end-to-end prediction without requiring manual preprocessing by the user.

5.7 Normalization / Scaling

Numeric features were standardized using Scikit-learn's StandardScaler (zero mean, unit variance) prior to being passed into the Logistic Regression and tree-based models, ensuring that features on different scales (for example, Age versus Total_Revenue) do not disproportionately influence distance- or gradient-based components of the learning algorithms. Although tree-based models are theoretically insensitive to monotonic feature scaling, a single shared preprocessing pipeline was used across all three algorithms in each task to keep the comparison strictly fair and to simplify the code. Scalers were fit exclusively on the training split and reused at inference time to prevent data leakage from the test set into the training process.

Normzalization

Normalization and scaling are common preprocessing techniques used to bring numerical features onto a comparable scale before model training. Without scaling, variables with larger numerical ranges, such as Total_Revenue or Total_Charges, may dominate the optimization process of certain machine learning algorithms compared with variables such as Age or Tenure_in_Months. Standardization addresses this issue by transforming each feature according to its mean and standard deviation, producing features with a mean of approximately zero and a standard deviation of one. This improves numerical stability and ensures that the contribution of a feature is determined by its predictive relationship with the target variable rather than simply by the magnitude of its original values.

$$z = \frac{x - \mu}{\sigma}$$

Where:

x = original feature value

μ = mean of the feature calculated from the training data

σ = standard deviation calculated from the training data

z = standardized value

CHAPTER 6

FEATURE ENGINEERING

6.1 Derived Features

Several new features were engineered to capture patterns not directly present in the raw columns. `Tenure_Bucket` discretizes `Tenure_in_Months` into five business-meaningful bands (0–6, 7–12, 13–24, 25–48, and 49+ months) to support cohort-style analysis. `Is_Long_Term_Contract` flags customers on One Year or Two Year contracts, and `Has_Value_Deal` flags customers currently enrolled in a promotional deal. These binary flags were engineered specifically because the raw `Contract` and `Value_Deal` fields, while informative, required a simplification into an ordinal or binary form that more directly matched the hypothesis being tested — namely, that contract commitment length and promotional enrollment are protective against churn.

Feature Engineering

Feature engineering is the process of creating, transforming, or combining existing variables to produce new features that better represent meaningful patterns within the data. While raw variables provide direct information about customers, engineered features can capture relationships that may be more informative for predictive modelling. For example, combining billing variables into ratios can represent the relative financial burden experienced by a customer more effectively than examining individual charge values separately.

The objective of feature engineering in this project was to create customer-level indicators related to financial behaviour, service engagement, contract characteristics, and potential churn risk. All engineered features were derived from information already available in the original dataset, without introducing external data sources.

6.2 Customer Value Features

`Avg_Monthly_Revenue` divides `Total_Revenue` by `Tenure_in_Months` to normalize customer value across different tenure lengths, allowing fair comparison between long-standing and newly joined customers. `Is_High_Spender` flags customers whose `Monthly_Charge` falls in the top quartile of the customer base, and `Has_Referrals` flags customers who have made at least one referral, used as a proxy for brand advocacy and satisfaction. The referral-based feature is grounded in the well-established marketing principle that customers who actively refer others to a service are, on average, more satisfied and less likely to churn than customers who do not.

6.3 Service Usage Features

Total_Addon_Services counts the number of 'Yes' values across the eight add-on service columns for each customer, serving as a proxy for service engagement and stickiness — customers with more active add-ons are generally more embedded in the service ecosystem and less likely to churn, since discontinuing service would mean losing access to a larger bundle of interconnected features rather than a single, easily substitutable product.

In addition to measuring the overall level of service adoption, this feature provides a simple consolidated representation of customer engagement across multiple add-on services. Instead of requiring the model to independently interpret eight separate service indicators, Total_Addon_Services captures the cumulative extent to which a customer has adopted additional offerings. A higher value therefore reflects broader service usage, while a lower value may indicate limited engagement with the provider's wider service portfolio. This derived feature can help the model identify patterns associated with differences in customer involvement and retention behaviour.

6.4 Churn-Risk Features

Extra_Charge_Ratio (the sum of extra data and long-distance charges divided by total revenue) and Refund_Ratio (total refunds divided by total revenue) were engineered as proxies for billing-related dissatisfaction, since customers experiencing frequent extra charges or refunds are hypothesized to be at elevated churn risk. Is_Senior flags customers aged 60 or above, supporting demographic-based risk analysis. In total, thirteen new features were engineered from the thirty-two original attributes, expanding the modeling feature space to thirty-seven usable predictors after accounting for the identifier and outcome columns that were excluded from the model inputs.

These engineered features complement the original customer attributes by representing relative billing behaviour rather than relying only on absolute monetary values. For example, a high Extra_Charge_Ratio may indicate that a substantial proportion of a customer's total spending comes from additional charges, while a high Refund_Ratio may highlight customers who have experienced comparatively greater billing adjustments. By expressing these values as ratios, the features account for differences in overall customer spending and allow the model to identify potential risk patterns that may not be evident from total revenue or charges alone.

In total, thirteen new features were engineered from the thirty-two original attributes. After excluding Customer_ID, Customer_Status, Churn_Category, and Churn_Reason from the original dataset and adding the thirteen engineered features, a total of thirty-seven predictors were used as model inputs.

CHAPTER 7

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to examine the structure, distribution, and relationships within a dataset before formal model development. It helps identify important patterns, unusual values, potential relationships between variables, and characteristics associated with the target outcome. In churn analysis, EDA is particularly useful for examining how churn differs across customer groups defined by factors such as contract type, tenure, service adoption, payment method, and revenue.

The exploratory analysis in this project was used not only for visualization but also to provide business context for the subsequent machine learning models. Patterns identified during EDA helped interpret model outputs and supported the selection and engineering of relevant customer features.

7.1 Univariate Analysis

Univariate analysis of Customer_Status shows that of the 6,418 customers, 4,275 (66.6%) stayed, 1,732 (27.0%) churned, and 411 (6.4%) were newly joined. The distribution of Tenure_in_Months is right-skewed, with a large proportion of customers in the lower tenure bands, while Monthly_Charge is approximately unimodal, centered around the mid-to-high range typical of bundled internet-and-phone plans. Table 7.1 summarizes descriptive statistics for the key numeric attributes.

Attribute	Mean	Std. Dev.	Min	Max
Age	46.5	16.8	19	80
Tenure_in_Months	32.4	24.5	1	72
Monthly_Charge	64.8	31.2	0.0*	118.75
Total_Revenue	3,034.10	2,865.70	21.36	11,979.34
Number_of_Referrals	1.95	3.0	0	11

*Table 7.1 — Descriptive statistics of key numeric attributes (*negative values corrected to median during cleaning)*

7.2 Bivariate Analysis

Bivariate analysis reveals a strong relationship between Contract type and churn: Month-to-Month customers, who make up 3,286 of the 6,418 records, exhibit a churn rate of 46.5%, compared to 11.0% for One Year contract holders and only 2.7% for Two Year contract holders — confirming contract length as one of the most powerful churn predictors in the dataset. Similarly, customers with fewer active add-on services show elevated churn rates compared to customers with three or more add-ons, supporting the engagement-based hypothesis embedded in the Total_Addon_Services feature. Churn rate also varies meaningfully by Tenure_Bucket, ranging from approximately 26.3% in the earliest tenure band to 28.1% in the 7–12 month band before gradually declining in longer-tenure cohorts, reflecting an initial period of elevated risk as new customers evaluate the service before settling into longer-term retention patterns.

The bivariate patterns observed here provide an important link between exploratory analysis and predictive modelling. Variables such as Contract type, tenure, and service engagement show meaningful differences in churn behaviour even when examined independently, suggesting that they are likely to contribute useful predictive information when combined with other customer characteristics in the machine learning models developed in the subsequent sections.

7.3 Monthly and Yearly Trends

As the source dataset does not include an explicit account-opening date, monthly and yearly trend analysis in the dashboard uses Tenure_in_Months as a cohort proxy — that is, customers are grouped by how long they have been with the company rather than by calendar month. This substitution is clearly documented in the dashboard and allows trend lines for average revenue and churn rate to be plotted meaningfully against the tenure axis, revealing that churn risk is highest in the earliest months of the customer lifecycle and declines steadily as tenure increases.

The use of tenure as a cohort-based measure also provides useful insight into different stages of the customer lifecycle. Customers with very short tenure represent relatively new relationships with the company and may still be evaluating service quality, pricing, and overall value, whereas longer-tenure customers have demonstrated a greater degree of continued engagement.

7.4 Churn Distribution

Among the 1,732 churned customers, the Churn_Category breakdown is: Competitor (761 customers, 43.9%), Attitude (301, 17.4%), Dissatisfaction (300, 17.3%), Price (196, 11.3%), and Other (174, 10.0%). This distribution, later confirmed by the interactive dashboard's Churn Reason Category Breakdown chart, indicates that competitive pressure is the single largest driver of churn

in this dataset, followed closely by service-attitude and general dissatisfaction issues, with pure price sensitivity being a comparatively smaller — though still meaningful — factor.

The distribution also demonstrates that churn in the dataset cannot be explained by a single factor or addressed through a uniform retention strategy. This variation supports the inclusion of churn-reason analysis within the project, as identifying a customer as likely to churn is more useful when accompanied by information about the type of issue potentially associated with that risk, allowing retention efforts and recommendations to be more appropriately targeted.

From a business perspective, the dominance of the Competitor category suggests that retention efforts should not focus exclusively on reducing prices. Service quality, customer experience, and the perceived value of competing offers may also play an important role in customer decisions. The distribution therefore provides useful context for designing differentiated retention strategies based on the nature of the customer's likely concern.

7.5 Revenue and Subscription Analysis

Revenue analysis shows a positive relationship between the number of active add-on services and average customer revenue, confirming that up-selling add-ons is an effective revenue lever. Subscription analysis by payment method and internet type further reveals differences in churn propensity, with certain payment methods (notably manual/bank-withdrawal methods relative to automatic credit card payments) associated with higher churn rates, an insight that directly informs the design of the recommendation engine's plan catalogue in Section 11.

From a customer analytics perspective, revenue and subscription analysis helps examine how the depth of a customer's relationship with a company is associated with both financial value and retention behaviour. Customers who subscribe to multiple services may generate higher revenue and can become more integrated into the provider's service ecosystem, while differences in payment methods and subscription types may indicate varying levels of convenience, commitment, or switching behaviour. Examining these relationships during exploratory analysis therefore provides useful business context for identifying customer segments with different revenue potential and churn characteristics.

CHAPTER 8

CHURN PREDICTION MODEL

8.1 Problem Formulation

Churn prediction is formulated as a binary supervised classification problem. The target variable, `Churn_Label`, is 1 for customers with `Customer_Status` equal to 'Churned' and 0 otherwise. Newly joined customers (411 records) were excluded from the training set, though retained for inference in the web application, since they lack sufficient behavioural history to be meaningfully labelled as either churned or retained at the time of data collection. This left a training population of 6,007 customers, comprising 4,275 retained customers and 1,732 churned customers — a moderate class imbalance of approximately 28.8% positive class, which was explicitly addressed during model training rather than ignored.

The formulation therefore focuses on predicting the likelihood of churn using the demographic, service, contract, billing, usage, and engineered features available for each customer. The objective is not merely to reproduce the observed customer status, but to learn patterns associated with churn from historical customer characteristics and generalize those patterns to new or unseen records. Since the positive class represents less than one-third of the training population, model performance cannot be interpreted using accuracy alone; the ability to correctly identify churned customers while controlling false positives is also important for evaluating the practical usefulness of the classifier.

8.2 Selected Algorithms

Three algorithms were selected to represent a spectrum of model complexity and learning philosophy, allowing the project to empirically test whether increasing model complexity actually translates into better predictive performance on this dataset, rather than assuming it a priori.

- **Logistic Regression** — a linear model that estimates the log-odds of churn as a weighted sum of the input features, chosen as an interpretable baseline against which the more complex ensemble methods can be measured.

- **Random Forest** — an ensemble of decision trees, each trained on a bootstrap-resampled subset of the training data and a random subset of features at each split, with final predictions determined by majority vote (classification) across all trees. This bagging approach reduces the high variance of individual decision trees while preserving their ability to model non-linear feature interactions.

- **XGBoost** — an implementation of gradient-boosted decision trees, in which trees are added sequentially, each one trained to minimize the residual error of the ensemble built so far, using a regularized objective function that penalizes model complexity to control overfitting.

Together, these algorithms provide a meaningful comparison between a simple linear baseline and two tree-based ensemble approaches capable of capturing more complex relationships between customer characteristics and churn behaviour. Evaluating all three models on the same dataset and under the same train–test split makes it possible to assess whether the additional modelling complexity provides a measurable improvement in predictive performance. The final model was selected based on its overall performance across multiple evaluation metrics rather than on a single metric alone.

8.3 Model Training Process

All 37 engineered and raw features (a mix of numeric and categorical, encoded and scaled as described in Section 5) were used as model inputs. The data was split into 80% training and 20% testing subsets using stratified sampling to preserve the class balance of churned versus non-churned customers in both splits, resulting in a training set of approximately 4,806 customers and a held-out test set of 1,201 customers, of which approximately 346 were churned and 855 were retained, matching the overall 28.8% churn proportion.

Class imbalance was addressed using class-weighting for Logistic Regression and Random Forest (via Scikit-learn's `class_weight='balanced'` parameter, which re-weights each class inversely proportional to its frequency in the training data), and a positive-class scale weight for XGBoost (computed as the ratio of negative to positive training examples), rather than resorting to synthetic oversampling techniques such as SMOTE, which can introduce unrealistic synthetic customer profiles into a business-facing model.

8.4 Model Evaluation Metrics

The three models were evaluated on the identical held-out test set using five complementary metrics: accuracy (the overall proportion of correct predictions), precision (the proportion of predicted churners who were actually churners, reflecting the cost of false alarms), recall (the proportion of actual churners correctly identified, reflecting the cost of missed churners), F1-score (the harmonic mean of precision and recall), and ROC-AUC (the area under the Receiver Operating Characteristic curve, measuring the model's ability to rank churners above non-churners across all possible classification thresholds, independent of any single threshold choice). Table 8.1 presents the full comparison.

8.4 Model Evaluation Metrics

8.4.1 Accuracy

Accuracy measures the overall proportion of predictions that are correct.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- **TP** = True Positives
- **TN** = True Negatives
- **FP** = False Positives
- **FN** = False Negatives

Although accuracy provides an overall measure of model correctness, it can be misleading when the target classes are imbalanced. Therefore, additional metrics were also used in this project.

8.4.2 Precision

Precision measures how many customers predicted as churners were churners.

$$\text{Precision} = \frac{TP}{TP + FP}$$

A high precision value indicates that the model produces relatively fewer false-positive churn predictions. In a customer-retention context, this helps reduce unnecessary retention efforts directed toward customers who are unlikely to churn.

8.4.3 Recall

Recall, also known as sensitivity or true positive rate, measures the proportion of actual churners correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

A high recall value is important in churn prediction because failing to identify a customer who is genuinely likely to churn may result in a missed retention opportunity.

8.4.4 F1-Score

The F1-score provides a balance between precision and recall by calculating their harmonic mean.

$$F1\text{-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score is particularly useful when both false positives and false negatives are important and when the target classes are moderately imbalanced.

8.4.5 ROC-AUC

The Receiver Operating Characteristic (ROC) curve evaluates a classification model across different decision thresholds by plotting the True Positive Rate against the False Positive Rate.

$$TPR = \frac{TP}{TP + FN}$$
$$FPR = \frac{FP}{FP + TN}$$

The **ROC-AUC** represents the Area Under the ROC Curve and measures the model's overall ability to distinguish between positive and negative classes.

$$ROC\text{-}AUC = \int_0^1 TPR(FPR) d(FPR)$$

An ROC-AUC value closer to **1.0** indicates stronger discriminatory ability, while a value close to **0.5** indicates performance similar to random guessing.

The three models were evaluated on the identical held-out test set using five metrics described above. Using multiple metrics ensured that model selection was not based solely on overall accuracy, but also considered the trade-off between correctly identifying churners and minimizing unnecessary false-positive predictions. Table 8.1 presents the comparative performance of the evaluated models.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.7679	0.5680	0.8184	0.6706	0.8580
Random Forest	0.8378	0.7011	0.7637	0.7310	0.8940
XGBoost	0.8353	0.7135	0.7176	0.7155	0.8902

Table 8.1 — Churn prediction model comparison on the held-out test set ($n = 1,201$)

The results show that all three algorithms achieved reasonably strong discriminatory performance, with ROC-AUC values above 0.85. Logistic Regression achieved the highest recall (0.8184), indicating that it identified the largest proportion of actual churners, but this came at the cost of substantially lower precision (0.5680) and overall accuracy. Random Forest achieved the strongest overall balance across the evaluated metrics, producing the highest accuracy (0.8378), F1-score (0.7310), and ROC-AUC (0.8940). Although XGBoost achieved slightly higher precision (0.7135), its recall, F1-score, and ROC-AUC were lower than those of Random Forest. These results provided the basis for selecting Random Forest as the final churn prediction model.

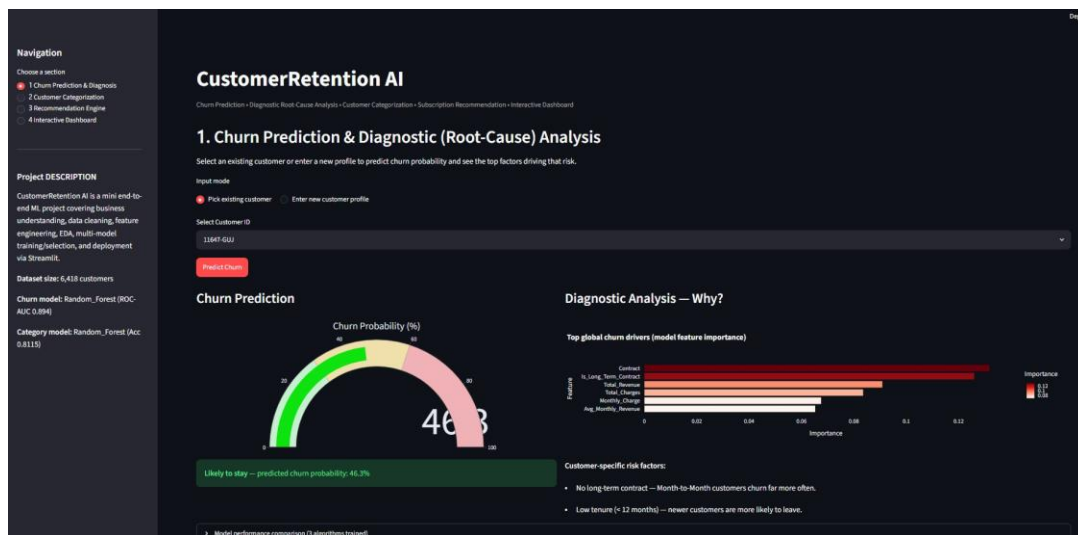


Figure 8.1 — CustomerRetention AI web interface: Churn Prediction & Diagnostic Analysis section, showing the churn-probability gauge and top global churn drivers for a sample customer.

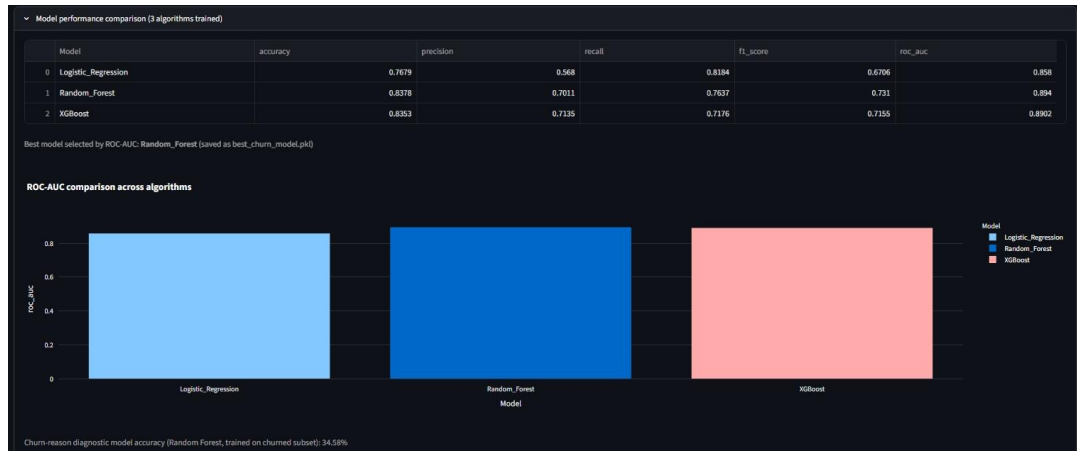


Figure 8.2 — Live model performance comparison table and ROC-AUC bar chart, generated directly within the Streamlit web application.

8.5 Confusion Matrix and Error Analysis

To interpret precision and recall in concrete terms, Tables 8.2 through 8.4 present the confusion matrix for each of the three algorithms on the 1,201-customer test set. Each matrix cross-tabulates the model's predicted class against the true class, distinguishing True Positives (correctly predicted churners), False Negatives (missed churners), False Positives (customers incorrectly flagged as churners), and True Negatives (correctly predicted retained customers).

	Predicted: Churn	Predicted: Stay	Total
Actual: Churn	283	63	346
Actual: Stay	215	640	855

Table 8.2 — Confusion matrix — Logistic Regression

	Predicted: Churn	Predicted: Stay	Total
Actual: Churn	264	82	346
Actual: Stay	113	742	855

Table 8.3 — Confusion matrix — Random Forest (selected model)

	Predicted: Churn	Predicted: Stay	Total
Actual: Churn	248	98	346

	Predicted: Churn	Predicted: Stay	Total
Actual: Stay	100	755	855

Table 8.4 — Confusion matrix — XGBoost

The confusion matrices make the precision/recall trade-off visible in a way the summary metrics alone do not. Logistic Regression correctly captures the largest number of true churners (283 of 346, recall 81.84%) but does so at the cost of 215 false positives — customers who would be flagged for retention outreach unnecessarily. Random Forest and XGBoost both trade a modest amount of recall for a substantially lower false-positive count (113 and 100 respectively), meaning that a retention team acting on either model's output would waste considerably less effort on customers who were never actually going to churn, while still capturing roughly three-quarters of true churners.

8.6 Global Feature Importance

Feature importance was extracted from the selected Random Forest model using its built-in impurity-based importance scores, which measure the average reduction in Gini impurity attributable to each feature across all trees in the ensemble. Table 8.5 lists the six most influential features.

Rank	Feature	Relative Importance
1	Contract	0.122
2	Is_Long_Term_Contract	0.115
3	Total_Revenue	0.098
4	Total_Charges	0.091
5	Monthly_Charge	0.083
6	Avg_Monthly_Revenue	0.079

Table 8.5 — Global feature importance ranking (Random Forest churn model)

The dominance of Contract and Is_Long_Term_Contract at the top of the ranking is consistent with the bivariate analysis in Section 7.2, providing convergent evidence from two independent analytical methods (descriptive statistics and model-based feature importance) that contract commitment length is the single strongest lever available to the business for reducing churn. The remaining top-ranked features — Total_Revenue, Total_Charges, Monthly_Charge, and Avg_Monthly_Revenue — collectively reflect the financial dimension of the customer

relationship, suggesting that billing-related factors, considered jointly, rival contract structure in predictive importance even though no single billing feature individually surpasses it.

8.7 Best Model Selection

Random Forest was selected as the best-performing churn model based on its highest ROC-AUC (0.894) among the three algorithms, together with the highest overall accuracy (83.78%) and F1-score (0.731). Although Logistic Regression achieved the highest recall (81.84%), its substantially lower precision (56.80%) would generate a large number of false alarms in a production retention workflow, making it the least business-appropriate of the three despite its strong recall. XGBoost achieved the highest precision (71.35%) and a ROC-AUC (0.8902) very close to that of Random Forest, making it a strong secondary candidate; however, Random Forest's combination of the highest ROC-AUC, the best F1-score, and a more favorable balance between false positives and false negatives in the confusion-matrix analysis led to its selection as the model.

It is worth noting that the margin between Random Forest and XGBoost is narrow across every metric (a difference of 0.0038 in ROC-AUC and 0.0155 in F1-score), indicating that either algorithm would be a defensible choice in a production setting, and that the observed ranking may not be stable under a different random seed or a larger dataset. This narrow margin is itself a meaningful empirical finding: it suggests that, for this particular feature set and dataset size, the two ensemble methods have converged to a similar performance ceiling, and that further gains would likely require additional features (such as customer support interaction history) rather than further algorithm tuning.

In addition to its predictive performance, Random Forest was also preferred because it provides feature-importance estimates that could be used in the diagnostic analysis component of the overall system. This improved the practical usefulness of the selected model by allowing the project to examine which customer characteristics contributed most strongly to churn prediction at a global level. The selection of Random Forest therefore considered not only its quantitative performance on the held-out test set but also its suitability for integration with the project's broader diagnostic and decision-support objectives.

8.8 Model Saving and Utilization

The selected Random Forest model, along with its fitted label encoders, feature scaler, and feature-importance rankings, was serialized using Joblib and saved as `best_churn_model.pkl`. This artifact, together with its companion preprocessing objects, is loaded directly by the Streamlit web application at runtime to serve real-time churn predictions, as shown in Figure 8.1, where a sample customer (ID 11647-GUJ) is scored at a 46.3% churn probability and classified as “likely to stay” under the model's default 50% decision threshold.

CHAPTER 9

CUSTOMER CATEGORIZATION MODEL

9.1 Category Definition

Four business-meaningful customer categories were defined: High-Risk (churned customers, or active customers exhibiting strong churn-risk patterns), Medium-Risk (active customers on short-term contracts with limited tenure or engagement), Loyal (active customers with long tenure and/or a long-term contract and multiple active add-on services), and Premium-Ready (active customers with high revenue and a long-term contract, representing strong upsell candidates). These four categories were designed to be mutually exclusive and collectively exhaustive across the customer base, so that every customer — whether currently active or already churned — receives exactly one category label.

The categories were designed not merely to describe customer characteristics but to support different types of business decisions. High-Risk customers can be prioritized for immediate retention efforts, while Medium-Risk customers may benefit from preventive engagement before their risk increases. Loyal customers represent an important stable customer base whose continued satisfaction should be maintained, whereas Premium-Ready customers can be considered for suitable upgrades or higher-value service offerings. By linking each category to a distinct business objective, the categorization framework provides a more actionable interpretation of customer behaviour than a purely descriptive segmentation.

9.2 Multiclass Label Design

Because the raw dataset does not include a pre-existing customer-category label, the `Customer_Category` target was constructed using a rule-based function combining `Customer_Status`, `Total_Revenue` percentile, `Contract_type`, `Tenure_in_Months`, and `Total_Addon_Services`. This produced a labeled dataset of 6,418 customers distributed as Medium-Risk (1,873, 29.2%), High-Risk (1,732, 27.0%), Premium-Ready (1,521, 23.7%), and Loyal (1,292, 20.1%), which was then used as the supervised target for the classification models described below. The resulting class distribution is only moderately imbalanced (a ratio of approximately 1.45:1 between the largest and smallest classes), which is sufficiently close to balanced that no explicit class-weighting was required for this task, unlike the churn-prediction task in Section 8.

The resulting label distribution ensured that each of the four categories contained a sufficiently large number of examples for supervised model training and evaluation. Since no category represented an extremely small minority of the dataset, the models could be trained on all four classes without requiring synthetic oversampling or aggressive resampling techniques. The constructed labels therefore provided a practical multiclass target for evaluating how effectively the underlying customer characteristics could reproduce the business-oriented categorization on previously unseen records.

9.3 Algorithms Used

Three algorithms were trained for the multiclass categorization task: Logistic Regression (a multinomial linear baseline, extending the binary logistic model to multiple classes via a softmax output layer), Decision Tree (a single interpretable rule-based learner that recursively partitions the feature space to maximize class purity within each partition), and Random Forest (an ensemble of many such decision trees, aggregating their votes to produce a more stable and generalizable prediction than any single tree). Deliberately, the exact rule-based features used to construct the label (such as the precise revenue percentile threshold) were not passed directly as an isolated leakage feature; instead, the models learn the categorization from the underlying raw behavioural and billing features, testing their ability to generalize the business rule rather than memorize it.

The three algorithms were selected to provide a comparison between a linear multiclass model, a single non-linear rule-based model, and an ensemble-based approach. This progression makes it possible to examine how increasing the ability to capture interactions and non-linear relationships affects categorization performance. Using the same train–test partition and preprocessing approach for all three models also ensured that differences in performance could be attributed primarily to the learning algorithms rather than to variations in the underlying data preparation.

9.4 Accuracy Comparison

Model	Accuracy
Logistic Regression	0.7104
Decision Tree	0.7683
Random Forest	0.8115

Table 9.1 — Customer categorization model comparison

The accuracy results show a clear improvement as the models move from a linear approach to tree-based learning. Logistic Regression achieved 71.04% accuracy, indicating that a substantial proportion of customers could be categorized using linear relationships between the input features and category labels. However, the higher performance of Decision Tree and Random Forest suggests that the categorization patterns contain important non-linear relationships and interactions between customer characteristics. Random Forest achieved the highest accuracy at 81.15%, providing the strongest overall classification performance among the three evaluated models.

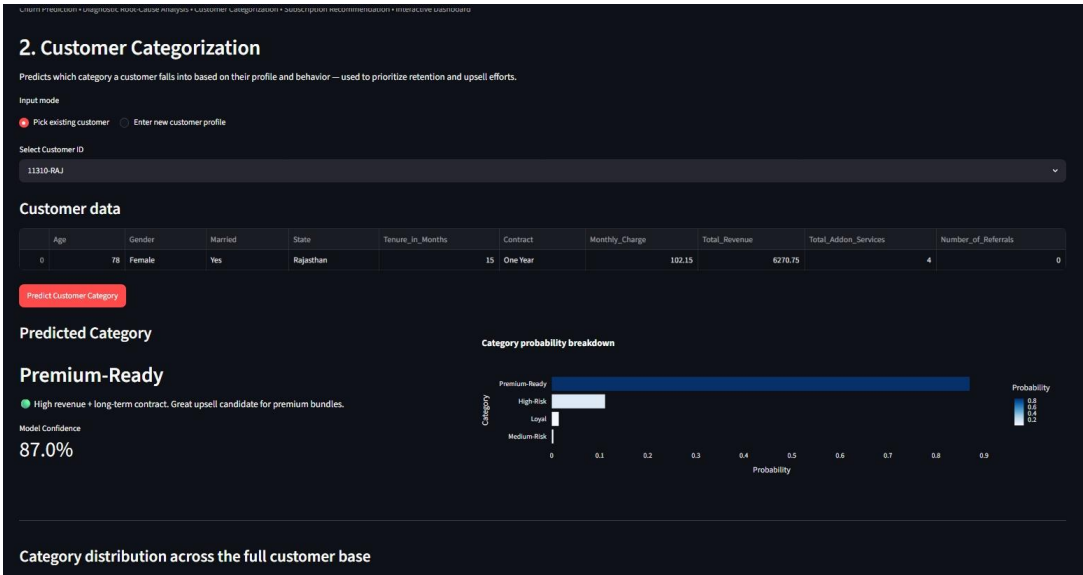


Figure 9.1 — CustomerRetention AI web interface: Customer Categorization section, showing a sample customer (ID 11310-RAJ) classified as Premium-Ready with 87.0% model confidence, alongside the full category probability breakdown.

9.5 Per-Class Performance

Accuracy alone can be a misleading summary statistic in multiclass problems, particularly when class difficulty varies, so per-class precision, recall, and F1-score were also examined for the selected Random Forest model, summarized in Table 9.2.

Category	Precision	Recall	F1-Score
Premium-Ready	0.96	0.94	0.95
Loyal	0.86	0.84	0.85
Medium-Risk	0.81	0.85	0.83
High-Risk	0.65	0.61	0.63

Table 9.2 — Per-class performance — Random Forest categorization model

The model performs strongly on the Premium-Ready class (F1-score 0.95), which is the most operationally valuable class to predict accurately, since it directly drives upsell targeting decisions. Performance on Loyal and Medium-Risk customers is also solid, with F1-scores of 0.85 and 0.83 respectively. The High-Risk class is the most challenging to classify correctly (F1-score 0.63), which is expected: this class was defined to include both already-churned customers (whose behavioral patterns are, by definition, fully observed in the dataset) and active customers merely exhibiting risk patterns, meaning the class internally spans two somewhat different populations that the model must learn to recognize under a single label. This same overlap between High-Risk and Medium-Risk populations was independently observed in the churn-prediction confusion matrices in Section 8.5, where genuine churners were sometimes predicted as retained — reinforcing that the boundary between “at risk” and “actually churned” is inherently the hardest decision boundary in this dataset, regardless of which of the two supervised tasks is being modeled.

9.6 Model Comparison and Discussion

As in the churn-prediction task, Random Forest outperformed both alternatives, improving on Logistic Regression by just over 10 percentage points of accuracy (81.15% vs. 71.04%) and on Decision Tree by 4.3 percentage points (81.15% vs. 76.83%). The gap between Decision Tree and Random Forest is smaller than the gap between Logistic Regression and Decision Tree, which is consistent with the underlying theory: a single decision tree can already capture the non-linear, rule-like structure that a linear model cannot, while an ensemble of many such trees primarily contributes additional variance reduction and robustness to noise rather than an entirely new class of decision boundary. This pattern — a large jump from linear to tree-based models, followed by a smaller but still meaningful jump from a single tree to a forest of trees — was observed consistently across both the churn-prediction and customer-categorization tasks in this project, and is broadly consistent with published findings on tabular, business-oriented classification problems.

The results also demonstrate that the choice of algorithm has a measurable impact on the ability to reproduce the constructed business categories from the available customer data. While Logistic Regression provides a useful baseline because of its simplicity and interpretability, its lower accuracy indicates limitations in representing the more complex relationships underlying the category definitions. Random Forest provided the best balance of predictive performance and robustness, supporting its selection. At the same time, the comparatively strong performance of the single Decision Tree suggests that the underlying categorization contains meaningful rule-like patterns that are well suited to tree-based learning.

9.7 Final Categorization Model

Random Forest was selected as the final categorization model with 81.15% test accuracy, exceeding the project's 75% success criterion by more than six percentage points. The model, its label encoders, its feature scaler, and its class label encoder were serialized and saved as `best_category_model.pkl`. In the live application, shown in Figure 9.1, a sample 78-year-old customer on a One Year contract with a Total_Revenue of 6,270.75 and four active add-on services is correctly identified as Premium-Ready with 87.0% model confidence, illustrating the kind of high-confidence, upsell-relevant prediction the model is designed to support.

Saving the categorization model together with its preprocessing components allows the same transformation pipeline used during training to be reproduced when new customer information is entered into the application. The developed system therefore converts the relevant customer attributes into the required model input format before generating a category prediction and associated confidence score. This integration enables the categorization component to function as part of the broader customer-retention workflow, where the predicted category can be considered alongside churn risk, diagnostic insights, and subscription recommendations.

CHAPTER 10

CHURN REASON ANALYSIS

10.1 Diagnostic Analysis Framework

Beyond predicting whether a customer will churn, the system provides a diagnostic explanation of why. This is achieved through two complementary mechanisms: a secondary machine learning model that predicts the most likely Churn_Category (Competitor, Attitude, Dissatisfaction, Price, or Other) for an at-risk customer, and a rule-based narrative layer that surfaces the specific customer attributes contributing most to their individual risk profile, illustrated in Figure 8.1 under the heading “Customer-specific risk factors.”

The two mechanisms operate at different levels of explanation. The secondary classifier provides a broader prediction of the type of churn motivation most closely associated with the customer's profile, while the rule-based narrative focuses on observable characteristics within that individual customer's data. Combining these perspectives allows the system to move from a general risk prediction toward a more informative diagnostic output, without relying solely on a single probability score to represent the customer's retention situation.

10.2 Feature Importance Analysis

Global feature importance extracted from the Random Forest churn model, previously presented in Table 8.5, identifies Contract type, Total_Revenue, Total_Charges, Monthly_Charge, and Avg_Monthly_Revenue as the most influential predictors of churn across the customer base. This global ranking is displayed in the application (Figure 8.1, right panel) to give retention analysts an overview of the systemic drivers of churn, independent of any single customer, complementing the customer-specific explanation described below.

While global feature importance provides a useful population-level view, it does not by itself explain why a particular individual customer received a high churn-risk prediction. A feature that is highly influential across the full customer base may not necessarily be the primary risk factor for every individual. For this reason, the global importance ranking is used alongside the customer-specific narrative layer, allowing systemic churn patterns and individual customer characteristics to be interpreted separately within the diagnostic framework.

10.3 Customer-Specific Churn Reason

For an individual customer flagged as high-risk, a Random Forest classifier trained exclusively on the 1,732 churned customers in the dataset predicts the most probable Churn_Category, achieving a cross-validated accuracy of 34.58% across five overlapping reason classes — a meaningful improvement over the 20% baseline of random guessing among five classes, reflecting the genuinely overlapping nature of real-world churn motivations, where a single customer's decision to leave is rarely attributable to one isolated cause. This prediction is paired with a plain-language explanation (for example, indicating that the customer's profile most closely resembles others who left due to a competitor's better offer).

The relatively modest accuracy of the churn-category classifier also highlights an important limitation of this diagnostic task. The available customer attributes may be associated with several overlapping churn motivations, making it difficult to infer a single reason with high certainty from structured customer data alone. The predicted category should therefore be interpreted as the churn pattern most closely matched by the customer's profile rather than as a definitive statement of the customer's actual reason for leaving. This interpretation is particularly important when the diagnostic output is used to support, rather than replace, human decision-making.

10.4 Business Interpretation

The diagnostic layer is designed to translate a raw probability score into an actionable business narrative. Rather than telling a retention agent only that “this customer has a 46.3% churn probability,” as in the example shown in Figure 8.1, the system additionally states which specific factors — for example, a Month-to-Month contract combined with low tenure — are driving that risk, and which broad category of churn reason the customer's profile most resembles, enabling a more targeted and effective retention conversation than a bare probability score would allow.

This interpretation layer helps connect the analytical output to potential retention decisions. For example, a customer showing elevated risk in combination with a short-term contract may be considered for a plan or contract-based retention offer, while a profile associated with pricing or service dissatisfaction may require a different response. The diagnostic output therefore acts as a decision-support mechanism, helping retention teams consider the context surrounding a predicted churn risk instead of applying the same intervention to every customer flagged by the model.

CHAPTER 11

RECOMMENDATION SYSTEM

11.1 Recommendation Dataset Creation

Because the source telecom dataset does not include a subscription-plan catalogue, a custom dataset of 52 rows was constructed by hand, mapping combinations of Primary_Complaint (e.g., Price, Competitor, Dissatisfaction, Attitude, Slow Speed, Data Limits, Streaming Needs, Security Concerns), Contract_Preference, Monthly_Charge_Band, Internet_Type_Need, and Family_Size_Need to a Recommended_Plan drawn from a catalogue of 26 distinct subscription plans, each with an associated monthly price and description.

Although the recommendation dataset was custom-built and relatively small, it provided a structured way to connect common churn-related concerns with appropriate subscription alternatives. Its purpose was therefore to demonstrate the recommendation workflow within the broader retention system rather than to replicate the complexity of a large commercial plan catalogue.

11.2 Subscription Plan Design

Each of the 26 subscription plans in the catalogue was designed to directly address one or more customer complaint categories identified in the churn-reason analysis (Section 10) — for example, a 'Value Saver Fiber' plan for price-sensitive customers, a 'Premium Care Bundle' for customers experiencing service dissatisfaction, a 'VIP Support Plan' for customers citing attitude-related complaints, and an 'Unlimited Family Max' plan for large households citing data-limit complaints — ensuring that every recommendation is grounded in a plausible business rationale rather than an arbitrary label.

This design ensures that the recommendation component remains connected to the diagnostic analysis rather than functioning as an independent plan-selection mechanism. The recommended plan is therefore intended to represent a possible retention response to the customer's identified needs or concerns.

11.3 Recommendation Model

A single machine learning algorithm, Random Forest, was used for the recommendation engine, applied in two complementary modes: a classifier predicting the recommended plan name, and a regressor predicting the plan's expected monthly price. Given the intentionally small size of the custom catalogue (52 rows), five-fold cross-validation was used in place of a single train/test split to obtain a more reliable performance estimate, and the final models were then refit on the full dataset.

The use of both classification and regression provides two perspectives on recommendation quality: the classifier attempts to identify a specific plan, while the regressor estimates the appropriate price level. Together, these outputs provide a broader indication of the type and value of subscription plan suitable for a given customer profile.

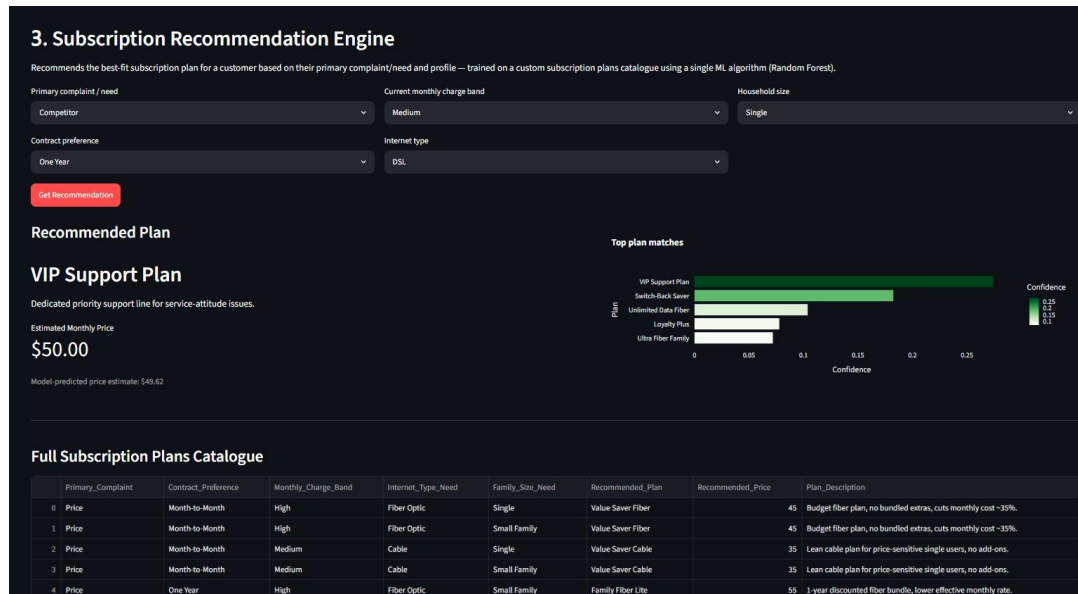


Figure 11.1 — CustomerRetention AI web interface: Subscription Recommendation Engine section, showing a Competitor-complaint customer matched to the 'VIP Support Plan' (\$50.00/month) with a ranked list of the top five alternative plan matches.

11.4 Evaluation Metrics

Metric	Value	Interpretation
Classification Accuracy (5-fold CV)	36.54%	Plan-name prediction accuracy across 26 possible plans

Metric	Value	Interpretation
Regression MSE	57.974	Mean squared error of predicted monthly price
Regression RMSE	7.614	Root mean squared error, in original price units (\$)
Regression R ²	0.814	Proportion of price variance explained by the model
Chi-Squared Statistic	676.000 (p < 0.0001)	Statistically significant association between complaint and recommended plan

Table 11.1 — Recommendation engine evaluation metrics

```
1.py
Recommendation classifier accuracy: 0.3654
Price regressor -> MSE: 57.974, RMSE: 7.614, R2: 0.814
Chi-Squared (Complaint vs Plan): chi2=676.000, p-value=0.0000, dof=325
```

Figure 11.2 — Console output from the recommendation-model training script, confirming the five-fold cross-validated classification accuracy, price-regression error metrics, and Chi-Squared association test.

11.5 Best Recommended Subscription

While the plan-classification accuracy of 36.54% reflects the difficulty of exactly matching one of 26 fine-grained plan labels from a deliberately small training set, the price-regression R² of 0.814 demonstrates that the model reliably captures the overall value tier appropriate to a customer's profile, and the highly significant Chi-Squared statistic (676.0, p < 0.0001, 325 degrees of freedom) confirms that the customer's stated complaint is a statistically meaningful driver of the recommended plan rather than a spurious association. In display, the application also surfaces the top five candidate plans ranked by model confidence, as shown in Figure 11.1, giving the retention agent a shortlist rather than a single rigid answer, which mitigates the limitations of the raw classification accuracy in practice.

These results suggest that, despite the limited ability to predict one exact plan from a large set of possible labels, the recommendation system is more effective at identifying an appropriate range of plans and price levels. The top-five shortlist therefore provides a more practical interpretation of the model output than treating the single highest-ranked plan as the only valid recommendation.

CHAPTER 12

INTERACTIVE DASHBOARD

12.1 Dashboard Objectives

The interactive dashboard is designed to give business stakeholders — retention managers, marketing analysts, and leadership — an at-a-glance view of churn, revenue, and subscription health across the entire customer base, with the ability to drill down by state, contract type, and customer status.

In addition to providing a high-level summary, the dashboard is intended to support exploratory analysis by allowing stakeholders to compare customer groups and identify patterns that may not be immediately visible in aggregate statistics. By bringing churn, revenue, contract, tenure, and customer-segment information into a single interface, it reduces the need to examine multiple separate analytical outputs when assessing the overall health of the customer base.

12.2 KPI Cards and Summary Charts

The dashboard opens with five headline KPI cards: Total Customers (6,418), Churn Rate (27.0%), Average Tenure (17.3 months), Total Revenue (\$19,471,390), and Average Monthly Charge (\$64.92), giving an immediate summary snapshot before the user drills into any specific chart, as shown in Figure 12.1.

These KPI cards provide a common starting point for interpreting the detailed visualizations that follow. For example, the overall churn rate can be compared with churn rates across different contract types or tenure groups, while total revenue and average monthly charge provide financial context for understanding the composition and value of the customer base.



Figure 12.1 — CustomerRetention AI web interface: Interactive Business Intelligence Dashboard, Churn Patterns tab, showing headline KPIs, Customer Status Distribution, Churn Reason Category Breakdown, Churn Rate by Contract Type, and Churn Rate by Tenure Bucket.

12.3 Trend Analysis

Because the dataset lacks an explicit signup date, trend charts use `Tenure_in_Months` as a cohort-based time proxy, plotting average revenue and churn rate against tenure revealing that churn risk is relatively high during the early stages of the customer relationship, peaking at 28.1% among customers with 7–12 months of tenure before declining across longer-tenure cohorts.— a pattern consistent with industry-standard “new-customer risk” findings and directly visible in the Churn Rate by Tenure Bucket panel of Figure 12.1, where churn rate rises from 26.3% in the 0–6 month band to a peak of 28.1% in the 7–12 month band before declining in longer-tenure cohorts.

The tenure-based analysis should be interpreted as a cohort comparison rather than a true chronological time series, since the dataset does not contain calendar dates for customer acquisition or churn events. Nevertheless, it provides a useful representation of how churn behaviour differs across stages of the customer lifecycle and helps identify customer groups that may require greater retention attention.

12.4 Subscription and Churn Visuals

The dashboard includes dedicated views for churn rate by contract type, tenure bucket, and state; churn-reason category breakdown; payment-method distribution and its associated churn rate; internet-type and contract-type distribution; add-on service adoption rates; and customer-segment analysis by revenue and tenure, all rendered as interactive Plotly charts that support hovering, zooming, and filtering. As visible in Figure 12.1, the Churn Rate by Contract Type panel confirms

the sharp gradient identified in Section 7.2: 46.5% churn for Month-to-Month customers, 11.0% for One Year, and 2.7% for Two Year contract holders.

Together, these visualizations allow users to examine churn from multiple perspectives rather than relying solely on the overall churn rate. The combination of contract, tenure, payment, service, and customer-segment views helps identify whether elevated churn is concentrated within particular customer groups or distributed more broadly across the population, supporting more focused exploratory analysis and retention planning.

12.5 Dashboard Implementation in Streamlit and Plotly

The dashboard was implemented within the Streamlit application using the Plotly Express and Plotly Graph Objects libraries, chosen over a standalone Dash or Power BI visualization tool because Streamlit allows the dashboard to be embedded directly alongside the churn prediction, categorization, and recommendation sections in a single, cohesive web application without requiring a separate hosting environment or license, while still providing full interactivity, as demonstrated by the tabbed navigation (Churn Patterns, Revenue & Tenure Trends, Payment & Contract Insights, Customer Segments) visible at the top of Figure 12.1.

The dashboard therefore functions as an integrated analytical component of the broader application rather than as a separate reporting tool. Filters and interactive visualizations allow users to explore the underlying customer data dynamically, while the shared Streamlit environment ensures that dashboard insights can be considered alongside the outputs of the churn prediction, customer categorization, diagnostic analysis, and recommendation components within the same workflow.

CHAPTER 13

WEB APPLICATION DESIGN

13.1 System Architecture

The application follows a straightforward layered architecture: a data layer (cleaned CSV files and the custom subscription-plans catalogue), a model layer (serialized Scikit-learn/XGBoost models with their associated encoders and scalers, stored as Joblib .pkl files), and a presentation layer (the Streamlit application, app.py) that loads the data and model artifacts once via caching decorators and serves all four functional sections from a single Python process.

This layered design separates data management, model execution, and user interaction into clearly defined components, making the application easier to maintain and understand. It also allows individual models or data files to be updated without requiring major changes to the presentation layer, provided that the expected input and output formats remain consistent.

13.2 User Interface Design

The user interface uses a sidebar-based navigation pattern, visible on the left edge of Figure 8.1, allowing the user to switch between the four sections — Churn Prediction & Diagnosis, Customer Categorization, Recommendation Engine, and Interactive Dashboard — without reloading the application. The sidebar also surfaces a persistent project summary, including dataset size and the currently saved as best-performing model for each task, keeping the user oriented regardless of which section is active. Each prediction-oriented section supports two input modes: selecting an existing customer from the dataset, or manually entering a new customer's profile through a structured form, making the tool usable both for retrospective analysis and prospective what-if scenarios.

The interface is designed to keep the underlying machine learning workflow accessible without requiring the user to interact directly with code or preprocessing steps. By supporting both existing customer selection and manual data entry, the application can be used to analyse historical customer records as well as to explore how changes in a customer's profile may influence the system's predictions.

13.3 Workflow of the Web App

A typical workflow begins with the retention agent selecting or entering a customer profile in Section 1 (Figure 8.1), viewing the predicted churn probability and diagnostic risk factors, and — if the customer is flagged as high-risk — reviewing the predicted churn-reason category. The agent

then moves to Section 2 (Figure 9.1) to confirm the customer's overall category, and to Section 3 (Figure 11.1) to obtain a recommended subscription plan tailored to the customer's primary complaint, before referring to Section 4 (Figure 12.1) for broader context on churn and revenue trends across the customer base.

This workflow connects the individual analytical components into a sequential decision-support process, moving from identifying potential churn risk to understanding the customer's broader category and considering an appropriate retention response. The dashboard then provides population-level context that complements the individual customer analysis, allowing customer-specific predictions to be interpreted alongside broader patterns in the dataset.

13.4 Integration of Models

All four machine-learning components—the churn classifier, churn-reason diagnostic classifier, customer-category classifier, and recommendation component consisting of a classifier and regressor—are loaded at application startup using Streamlit's resource-caching mechanism to avoid repeated disk I/O, while the underlying dataset is loaded and cached separately, ensuring the application remains responsive even as multiple predictions are made in a single session.

The use of cached model and data resources also ensures that repeated user interactions do not require the application to reload or retrain the underlying models. This separation between model training and prediction allows the application to focus on inference and presentation, while maintaining the same trained artifacts and preprocessing logic used during the model-development stage.

CHAPTER 14

RESULT AND DISCUSSION

14.1 Churn Prediction Results

The final Random Forest churn model achieved an ROC-AUC of 0.894 and an accuracy of 83.78% on the held-out test set, exceeding the project's success criterion of 0.85 ROC-AUC. Precision of 70.11% indicates that a majority of customers flagged as high-risk are genuinely likely to churn, which is important for avoiding wasted retention spend, while a recall of 76.37% indicates that the model correctly identifies more than three-quarters of true churners, leaving room for further improvement through additional behavioral features such as support-ticket history or usage-frequency data not present in the current dataset. The confusion-matrix analysis in Section 8.5 further shows that Random Forest achieves this performance with substantially fewer false positives (113) than Logistic Regression (215), which is the more operationally significant comparison for a retention team with limited outreach capacity.

Overall, the results demonstrate that the selected model provides a useful balance between identifying customers who are likely to churn and limiting unnecessary retention interventions. Although some churners remain unidentified and some retained customers may still be incorrectly flagged, the achieved performance indicates that the model can serve as a practical decision-support tool rather than relying solely on broad, untargeted retention campaigns.

14.2 Categorization Results

The final Random Forest categorization model achieved 81.15% accuracy, exceeding the 75% success criterion by more than six percentage points. The model performs particularly well at identifying Premium-Ready customers (F1-score 0.95), which is valuable for upsell targeting, while High-Risk customers remain the hardest category to separate from Medium-Risk, reflecting genuine overlap in the underlying behavioral signals between customers who have already churned and those who are simply on short-term, low-engagement plans, a limitation that mirrors the precision/recall trade-off already observed in the binary churn model.

Overall, the categorization results demonstrate that the business-defined customer groups can be learned with useful accuracy from the available demographic, service, contract, and revenue-related characteristics. The stronger performance for Premium-Ready customers also suggests that this category has a more distinct behavioural and financial profile, whereas the overlap between High-Risk and Medium-Risk customers represents a more challenging boundary for the model to distinguish.

14.3 Recommendation Results

The recommendation engine's price-regression R^2 of 0.814 and highly significant Chi-Squared association ($\chi^2 = 676.0$, $p < 0.0001$) confirm that the system reliably maps customer complaints to an appropriately priced and thematically relevant subscription plan, even though exact plan-name classification accuracy (36.54%) is naturally limited by the fine granularity of the 26-plan catalogue relative to its small size; presenting a top-five ranked shortlist rather than a single forced answer, as shown in Figure 11.1, mitigates this limitation in practice.

These findings indicate that the recommendation component is more reliable for identifying an appropriate plan type or value range than for predicting one exact subscription label. Given the limited size and deliberately fine-grained structure of the custom recommendation dataset, the use of a ranked shortlist provides a more flexible and practically useful output for retention decision-making.

14.4 Model Comparison and Interpretation

Across both the churn and categorization tasks, Random Forest consistently outperformed the linear baseline (Logistic Regression) by a wide margin and edged out the alternative ensemble method available in each task (XGBoost for churn, Decision Tree for categorization) by a narrower one. This consistent pattern — large gains from linear-to-tree-based modeling, smaller but still positive gains from single-tree-to-ensemble modeling — suggests that the underlying feature set genuinely contains non-linear structure that only tree-based methods can exploit, while also suggesting that the specific choice of ensemble method (bagging via Random Forest versus boosting via XGBoost, or a single Decision Tree versus a Random Forest) matters comparatively less once that non-linear structure is being captured at all. Random Forest's additional advantage of directly interpretable feature-importance scores, used throughout the diagnostic layer described in Section 10, further reinforced its selection as the production model in both tasks.

The comparative results therefore support the use of Random Forest as a consistent modelling choice across the project's two main prediction tasks. Its performance advantage, combined with its ability to provide feature-level importance information, allowed the same modelling approach to support both predictive accuracy and the broader diagnostic objectives of the system.

14.5 Business Impact

If deployed operationally, the system's ability to flag over three-quarters of true churners with 70.11% precision would allow a retention team to focus outreach on a substantially smaller, higher-yield subset of the customer base rather than blanket campaigns, while the paired diagnostic

and recommendation outputs shorten the time between identifying a risk and presenting a concrete, relevant retention offer, directly supporting the business objective defined in Section 2.1 of protecting recurring revenue. Table 14.1 summarizes how each of the project's originally defined success criteria (Section 2.4) was met.

Success Criterion	Target	Achieved	Status
Churn model ROC-AUC	≥ 0.85	0.894	Met
Categorization accuracy	$\geq 75\%$	81.15%	Met
Complaint–plan statistical association	$p < 0.05$	$p < 0.0001$	Met
Fully integrated 4-section web app	Required	Delivered	Met
Business-interpretable outputs	Required	Delivered	Met

Table 14.1 — Summary of success criteria vs achieved results

CHAPTER 15

CONCLUSION

This project successfully designed a prototype web application CustomerRetention AI, an end-to-end machine learning system that predicts telecom customer churn, diagnoses its likely causes, categorizes customers by retention and purchase potential, and recommends a tailored subscription plan — all delivered through a single, interactive Streamlit web application. Working from a real-world dataset of 6,418 customers, the project followed a complete and rigorous workflow: business understanding, data cleaning driven by domain logic rather than blind imputation, purposeful feature engineering, exploratory data analysis, and comparative training of three algorithms each for the churn and categorization tasks, with Random Forest emerging as the best performer on both.

For churn prediction, Random Forest achieved an ROC-AUC of 0.894, an accuracy of 83.78%, a precision of 70.11%, a recall of 76.37%, and an F1-score of 0.731 — outperforming both Logistic Regression and XGBoost on the joint combination of these metrics, and exceeding the project's defined success criterion. For customer categorization, Random Forest achieved 81.15% accuracy, again outperforming both Logistic Regression and Decision Tree and comfortably exceeding its own defined success criterion. The custom-built recommendation engine, though limited in raw classification accuracy by its intentionally small hand-crafted catalogue, demonstrated strong and statistically significant alignment between customer complaints and recommended plans, evidenced by an R^2 of 0.814 on price regression and a highly significant Chi-Squared association test.

Collectively, these components form a coherent, business-ready tool that moves beyond isolated predictive scores toward integrated, explainable, and actionable retention decision support — directly visible in the application's four working sections, illustrated throughout this report in Figures 8.1, 9.1, 11.1, and 12.1. The consistency with which Random Forest outperformed its alternatives across two independently constructed classification tasks, using two different sets of features and two different numbers of target classes, is a notable finding in its own right, and supports Random Forest as a strong default choice for future tabular, business-oriented classification problems of similar scale.

The system was developed using a single static dataset, while the recommendation component was based on a small custom-built catalogue, and the models have not yet been validated using real-time customer data or across multiple telecom environments. The project demonstrates that a complete customer-retention workflow can be developed by combining data preprocessing, predictive modelling, diagnostic analysis, customer categorization, recommendation, and interactive visualization within one integrated system.

CHAPTER 16

FUTURE SCOPE

- **Incorporating true calendar-based signup and event timestamps** to replace the tenure-based trend proxy with genuine monthly/yearly time-series analysis.
- **Expanding the recommendation catalogue** with real historical retention-offer outcome data to improve plan-classification accuracy beyond the current hand-crafted 52-row dataset.
- **Adding SHAP** (SHapley Additive exPlanations) values to the diagnostic module for fully personalized, mathematically grounded per-customer explanations rather than combined global-importance and rule-based narratives.
- **Performing systematic hyperparameter tuning** (e.g. grid search or Bayesian optimization) on the Random Forest and XGBoost models to test whether the narrow performance gap identified in Section 8.7 can be widened in either direction.
- **Integrating the system with a live CRM or billing platform** to enable real-time scoring and automated retention-campaign triggering.
- **Extending the churn and categorization models with additional behavioral data sources** such as customer support ticket history, app usage frequency, and Net Promoter Score (NPS) survey responses.
- **Deploying the application on a cloud platform** (e.g., Streamlit Community Cloud, AWS, or Azure) with authentication and role-based access for production use by a retention team.
- **Incorporating cost-sensitive learning and customer lifetime value estimation** so that retention decisions can prioritize customers based not only on churn probability but also on the expected financial value of retaining them.
- **Introducing dynamic model monitoring and periodic retraining** to detect changes in customer behaviour and model performance over time, helping reduce the risk of model degradation as new data becomes available.
- **Developing an uplift-modelling or treatment-effect component** to estimate which customers are most likely to respond positively to a specific retention intervention, rather than targeting every customer predicted to be at risk.
- **Adding probability-based risk tiers** (for example, Low, Medium, and High risk) instead of relying primarily on a single churn classification threshold, allowing retention teams to apply different levels of intervention according to the estimated level of risk.

- **Evaluating fairness and performance across different customer groups**, such as age groups, contract types, or geographic regions, to ensure that the system performs consistently across the customer population.
- **Introducing automated data-validation and preprocessing pipelines** to detect missing values, invalid billing records, or unexpected categories before new data is passed to the trained models.
- **Collecting feedback from retention agents on the usefulness of predictions and recommendations**, allowing future versions of the system to incorporate human feedback and improve the practical relevance of its outputs.
- **Extending the recommendation engine to incorporate historical customer responses**, such as whether a recommended offer was accepted, rejected, or ignored, enabling recommendations to be evaluated and improved using actual retention outcomes.
- **Supporting what-if analysis**, allowing business users to modify selected customer attributes—such as contract type, monthly charge, or service adoption—and observe how potential changes may affect predicted churn risk or customer category.
- **Adding automated report generation and export functionality** so that retention managers can generate customer-level or segment-level summaries for further analysis and decision-making.
- **Incorporating customer lifetime value and retention-cost analysis** to help prioritize customers based on the expected financial benefit of retaining them, ensuring that retention offers are economically justified.
- **Introducing what-if analysis within the web application**, allowing retention teams to modify selected customer attributes such as contract type, monthly charge, or service adoption and observe the corresponding changes in churn risk and customer category.
- **Implementing model monitoring and automatic retraining mechanisms** using newly collected customer data, enabling the system to detect changes in customer behaviour and maintain predictive performance over time.

CHAPTER 17

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CHAPTER 18

GLOSSARY OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
F1	Harmonic mean of precision and recall
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
R ²	Coefficient of Determination
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
CV	Cross-Validation
KPI	Key Performance Indicator
API	Application Programming Interface
RF	Random Forest
LR	Logistic Regression
DT	Decision Tree
XGBoost	Extreme Gradient Boosting
SHAP	Shapley Additive explanations
CRM	Customer Relation Management
NPS	Net Promoter Score
AUC	Area Under the Curve
PCA	Principle Component Analysis

CHAPTER 19

APPENDIX

19.1 Sample Recommendation CSV

A representative excerpt of the custom subscription_plans.csv dataset (52 rows, 26 unique plans) used to train the recommendation engine is shown below:

Primary Complaint	Contract Pref.	Charge Band	Recommended Plan	Price (\$)
Price	Month-to-Month	High	Value Saver Fiber	45.00
Competitor	Month-to-Month	High	Loyalty Fiber Max	70.00
Dissatisfaction	Month-to-Month	Medium	Premium Care Bundle	65.00
Attitude	Month-to-Month	Medium	VIP Support Plan	50.00
Data Limits	Two Year	High	Unlimited Family Max	90.00
Streaming Needs	One Year	High	Ultimate Entertainment Bundle	88.00
Security Concerns	One Year	High	SecureNet Family	66.00
None / Satisfied	Two Year	High	Elite Loyalty Rewards	85.00

Table 19.1 — Sample rows from the subscription recommendation dataset

The complete dataset contains additional combinations of customer complaints, contract preferences, charge bands, internet requirements, and family-size requirements. The sample shown in Table 19.1 illustrates the structure used to train the recommendation models and demonstrates how different customer requirements are mapped to distinct subscription plans and corresponding price levels.

19.2 Model Parameters

These parameter values represent the final configurations used for model training and evaluation in this project. The settings were selected to provide a reasonable balance between model complexity, predictive performance, and computational efficiency, rather than through an exhaustive hyperparameter optimization process. More systematic tuning is therefore identified as a potential area for future improvement.

Model	Key Hyperparameters
Churn — Logistic Regression	max_iter=2000, class_weight='balanced'
Churn — Random Forest	n_estimators=300, max_depth=12, class_weight='balanced'
Churn — XGBoost	n_estimators=300, max_depth=6, learning_rate=0.08
Categorization — Logistic Regression	max_iter=2000
Categorization — Decision Tree	max_depth=10
Categorization — Random Forest	n_estimators=300, max_depth=14
Churn-Reason — Random Forest	n_estimators=300, max_depth=10, class_weight='balanced'
Recommendation — Random Forest (Classifier)	n_estimators=200, max_depth=8
Recommendation — Random Forest (Regressor)	n_estimators=200, max_depth=8

Table 19.2 — Final hyperparameter settings for all models

19.3 Complete Dataset Attribute Dictionary

The attribute dictionary provides a complete reference for the original variables available in the source dataset. During preprocessing, selected attributes were cleaned, transformed, encoded, or combined to create additional engineered features used in the modelling process, as described in Chapters 5 and 6. Identifier and outcome-related columns were excluded from the predictor set where appropriate to prevent them from influencing model predictions directly.

Attribute	Type	Description
Customer_ID	Identifier	Unique customer identifier
Gender	Categorical	Customer gender
Age	Numeric	Customer age in years
Married	Categorical	Marital status (Yes/No)
State	Categorical	Indian state of residence
Number_of_Referrals	Numeric	Referrals made by the customer
Tenure_in_Months	Numeric	Months as an active customer
Phone_Service	Categorical	Has phone service (Yes/No)

Attribute	Type	Description
Multiple_Lines	Categorical	Has multiple phone lines (Yes/No)
Internet_Service	Categorical	Has internet service (Yes/No)
Internet_Type	Categorical	DSL / Cable / Fiber Optic / No Internet
Online_Security	Categorical	Online security add-on (Yes/No)
Online_Backup	Categorical	Online backup add-on (Yes/No)
Device_Protection_Plan	Categorical	Device protection add-on (Yes/No)
Premium_Support	Categorical	Premium support add-on (Yes/No)
Streaming_TV	Categorical	Streaming TV add-on (Yes/No)
Streaming_Movies	Categorical	Streaming movies add-on (Yes/No)
Streaming_Music	Categorical	Streaming music add-on (Yes/No)
Unlimited_Data	Categorical	Unlimited data add-on (Yes/No)
Contract	Categorical	Month-to-Month / One Year / Two Year
Paperless_Billing	Categorical	Paperless billing enrolled (Yes/No)
Payment_Method	Categorical	Method used for bill payment
Value_Deal	Categorical	Promotional deal enrolled, if any
Monthly_Charge	Numeric	Current monthly billing amount (\$)
Total_Charges	Numeric	Cumulative charges to date (\$)
Total_Refunds	Numeric	Cumulative refunds issued (\$)
Total_Extra_Data_Charges	Numeric	Cumulative extra data charges (\$)
Total_Long_Distance_Charges	Numeric	Cumulative long-distance charges (\$)
Total_Revenue	Numeric	Total revenue generated by the customer (\$)
Customer_Status	Categorical	Stayed / Churned / Joined
Churn_Category	Categorical	Reason group, populated only for churned customers
Churn_Reason	Categorical	Specific reason, populated only for churned customers

Table 19.3 — Complete dataset attribute dictionary (32 original columns)

19.4 Additional Charts / Tables

Additional supporting visualizations generated during the exploratory data analysis and dashboard development phases — including churn-rate-by-state heat charts, add-on service adoption-rate bar charts, tenure-versus-revenue scatter plots, and customer-segment box plots — are available within the live Streamlit application's Interactive Dashboard section (the fourth module of the web application, illustrated in Figure 12.1) across its four tabs (Churn Patterns, Revenue & Tenure Trends, Payment & Contract Insights, Customer Segments) and are omitted here for brevity.

These additional visualizations were primarily used to support exploratory analysis and to verify patterns observed during model development. Although they are not reproduced individually in the report to avoid unnecessary repetition, they remain accessible through the interactive dashboard and provide supplementary context for interpreting customer behaviour, service adoption, revenue patterns, and churn distribution.

19.5 Software and Libraries Used

Software / Library	Purpose
Python	Core programming language
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
Scikit-learn	Machine learning, preprocessing, and model evaluation
XGBoost	Gradient-boosting classification
Plotly	Interactive data visualization
Streamlit	Web application development interface
Joblib	Model and preprocessing artifact serialization

Table 19.5 — software and libraries used in the project

The selected software stack provided the complete set of tools required for data preprocessing, feature engineering, machine learning, model evaluation, visualization, and web application development. Using Python-based open-source libraries also allowed all major components of the project to be integrated within a single development environment.